Richeek Das

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Research Interests

Self Supervised Representation Learning, Robotic Perception, Multimodal Sensing, Event-based Vision

Education

University of Pennsylvania Philadelphia, USA
PhD in Computer and Information Science *2023 - 2028*
Advisor: Prof. Pratik Chaudhari

Indian Institute of Technology Bombay Mumbai, India
Bachelor of Technology with Honors in Computer Science and Engineering *2019 - 2023*
Advisor: Prof. Preethi Jyothi *GPA: 9.62/10.0*
Thesis: Code-switched Text Modelling for Natural Language Understanding
Received the **Research Excellence Award** for undergraduate thesis

Journal Papers

2. **Richeek Das**, Kostas Daniilidis, Pratik Chaudhari, **Fast Feature Field (F^3): A Predictive Representation of Events**, *under review at Science Robotics 2025*

1. **Richeek Das**, Aaron Jerry Ninan, Adithya Bhaskar, Ajit Rajwade, **Performance Bounds for LASSO under Multiplicative LogNormal Noise: Applications to Pooled RT-PCR testing**, *accepted at the Signal Processing Journal 2024*

Conference Papers

6. **Richeek Das**, Pratik Chaudhari, **Neurosim: A Fast Simulator for Neuromorphic Robot Perception**, *on arXiv 2025*

5. **Richeek Das**, Samuel Dooley, **Fairer and More Accurate Tabular Models Through NAS**, *accepted at the Algorithmic Fairness through the Lens of Time workshop of NeurIPS 2023*

4. **Richeek Das**, Sahasra Ranjan, Shreya Pathak, Preethi Jyothi, **Pretraining Techniques for Improved Code-switched Natural Language Understanding**, *accepted at the ACL 2023 (Association for Computational Linguistics)* - **ACL Outstanding Paper Award (top 1%)**

3. Alex Markham, **Richeek Das**, Moritz Grosse-Wentrup, **A Distance Covariance-based Kernel for Nonlinear Causal Clustering in Heterogeneous Populations**, *accepted at the CLeaR 2022 (1st conference on Causal Learning and Reasoning)*

2. Alexander Erlei, **Richeek Das**, Lukas Meub, Avishek Anand, Ujwal Gadiraju, **For What It's Worth: Humans Overwrite Their Economic Self-interest to Avoid Bargaining With AI Systems**, *accepted at the ACM CHI 2022 (Conference on Human Factors in Computing Systems)*

1. Ashish Tiwari, **Richeek Das**, Shanmuganathan Raman, **Exploring Deeper Graph Convolutions For Semi-Supervised Node Classification**, *accepted at the IEEE ICASSP 2022 (International Conference on Acoustics, Speech, and Signal Processing)*

Research and Work Experience

Fairer and More Accurate Tabular Models Through NAS

Abacus.AI | Guide: Dr. Samuel Dooley — *May 2023 - Jul 2023*

- Proposed a novel multi-objective NAS technique to utilize the implicit fairness in model architectures
- Performed extensive experiments over model search spaces and proved the existence of architectures that are inherently fair and accurate – enabling us to search for these models which surpass current SOTA debiasers

Neural Architecture Search Toolkit for Computer Vision

Sony AI, Japan | Guide: Takuya Narihira, Hsingying Ho — *May 2022 - Jul 2022*

- Proposed a novel Once-For-All based NAS framework for Semantic Segmentation in NNabla NAS, with plug-and-play features for building dynamic sub-networks with differing hardware-constraints without any re-training
- Adapted the method to DeepLabv3+ with dynamic sub-networks – 15% increase in MIoU and 2× latency reduction

Representation Learning for Event-based Vision

UPenn, GRASP Lab | Guide: Prof. Pratik Chaudhari, Prof. Kostas Daniilidis — *Jul 2024 - Sep 2025*

- Developed a mathematical argument and efficient algorithms for building representations event camera data—retains information about the structure and motion of the scene, robust to noise and event rates (480 Hz at HD resolution)
- SOTA performance on optical flow, segmentation and monocular depth on data from 3 platforms (car, quadruped robot, drone), different lighting (daytime, nighttime) and environments (indoors, outdoors, urban, off-road)

Extracting Scene Material Properties from Radar Signals

UPenn, WAVES Lab | Guide: Prof. Mingmin Zhao — *Aug 2023 - Jan 2024*

- Built hardware – a handheld device with synchronized iPad LiDAR, TI mmWave radar, camera and an onboard computer to capture data in real 3D indoor scenes with diverse materials and geometries
- Built software – a differentiable radar simulator to model wave propagation, antenna patterns and ray tracing in 3D scenes, and a novel SSL framework for extracting material properies using the physics of wave-material interactions

Real-time Multirotor Simulation Software

UPenn, GRASP Lab | Guide: Prof. Pratik Chaudhari — *Mar 2025 - Ongoing*

- A fast and modular simulation environment for multirotors with onboard IMUs, RGB, event, depth and semantic cameras. Playground to test real high-speed agile flight algorithms in simulation – control, planning, perception
- Built CUDA kernels for 10× event generation – efficient integration with Habitat (∼2700 FPS on RTX4090)

Code-Switched Natural Language Understanding

Google India + IIT Bombay, CSALT | Guide: Prof. Preethi Jyothi — *Jul 2022 - Jan 2023*

- Implemented intelligent masking strategies for MLM pretraining and built architectures specific to code-switched tasks with significant improvements in downstream Question Answering and Sentiment Analysis
- Built a generalized framework to adapt existing language translation models for low-resource code-switched text generation with multiple constraints: formality, politeness, toxicity, semantic similarity and more

Kernel methods for Non-linear Causal Clustering

Universität Wien, Neuroinformatics Lab | Guide: Prof. Moritz Grosse-Wentrup — *May 2021 - Oct 2021*

- Implemented a distance covariance-based kernel to measure the similarity between underlying nonlinear causal structures of samples – clustering according to their parent causal structures in heterogeneous populations
- Simulated causal datasets with non-linear relations to numerically evaluate the performance bounds of the distance covariance-based dependence contribution kernel and compare it with standard RBF and Polynomial kernels

Belief Elicitation on the Impact of AI Systems

TU Delft, Delft AI Labs | Guide: Prof. Ujwal Gadiraju — *May 2021 - Oct 2021*

- Implemented Binarized Scoring Rule based criterion for Belief Elicitation of user behaviour, presumptions and trust on the usage of Decision Support Systems (AI-System) for Algorithmic Bargaining
- Built and deployed a DRF backend, Angular frontend, PostgreSQL DB application coupled with Redis + Celery task management, on a Heroku + GitHub deployment pipeline to host 2700+ crowdsource submissions

Feature Gating for Deeper Graph Convolution Networks

IIT Gandhinagar, CVIG Lab | Guide: Prof. Shanmuganathan Raman — *Dec 2020 - Jun 2021*

- Introduced feature gating and formulated a heuristic to award importance scores to graph nodes and node features
- Proposed the use of identity mapping, a modified form of residual connection and feature gating to create deep GCN models which tackle oversmoothing and achieve SOTA results for semi-supervised node classification

Weighted LASSO under Multiplicative LogNormal Noise

IIT Bombay | *Guide: Prof. Ajit Rajwade* *Jan 2022 - Feb 2023*

- Theoretically justified and interpreted the use of Tapestry Pooling across hospitals for pooled RT-PCR group testing
- Proposed a novel Weighted LASSO algorithm with data-dependent weights to perform sparse signal recovery under multiplicative LogNormal noise (example Gaussian) – performance backed using simulations on real RT-PCR data

Selected Projects

Video from a Single Exposure Coded Snapshot Spring 2021

- Implemented a MATLAB solution for coded aperture compressive temporal imaging to recover a sequence of frames from a single coded-snapshot to achieve temporal gains in video acquisition without spatial compromise

Compressed Sensing in Tomographic Reconstruction Spring 2021

- Implemented compressed sensing solution for tomographic reconstruction of brain MRI scans with small number of measurement angles – performed coupled reconstruction assuming similarity of consecutive acquisitions

Cluster Monitoring and Alert System Spring 2022

- Built a web-app with DRF, Angular and InfluxDB – telegraf servers on host machines to perform cluster profiling and alert low-resource warnings in real-time using socket-servers, based on thresholds set by users

Branch Predictors for trace-based Simulators Autumn 2021

- Implemented branch predictors TAGE and L-TAGE in ChampSim, and performed extensive comparisons with Bi-modal and Hashed Perceptron on the metrics of tag width, MPKI, TAGE Table Size and history length

Compiler for C-like Language Spring 2022

- Constructed a compiler handling a subset of the C language, designing a recursive descent parser using lex and yacc – supports type inference, semantic checks and translation of AST to linear three-address codes

Extending xv6 Operating System Autumn 2021

- Extended the xv6 OS with syscalls for demand paged memory allocation and custom fork implementations. Implemented thread synchronization, semaphores and a simple linux-based disk emulated filesystem in C

Selected Achievements and Awards

- Received the **ACL 2023 Outstanding Paper Award**, awarded to the **top 1%** of the accepted papers. (2023)
- Received the **Paul S. Darnell** named CIS PhD fellowship at UPenn. (2023)
- Received the **Thomas Dooie, Class of 1974 Research Excellence Award** for B.Tech Thesis. (2023)
- Among the select few undergrads to attend the **Google Research Week** hosted in Bangalore, India (2023)
- Secured a rank of **497** in JEE Main and **544** in JEE Advanced among **1.2 million** candidates (2019)
- Received the **INSPIRE** scholarship, awarded to **top 1%** of the **80k+** students in the **ISC** Exam (2019)
- Received the **KVPY Fellowship** for securing **77th** rank out of **150k+** candidates in all of India (2018)
- Secured **4th rank in India** in **ICSE** out of **180k+** candidates (2017)

Teaching Experience

Learning in Robotics | *Prof. Pratik Chaudhari* Spring 2026
Introduction to Machine Learning | *Prof. Biplab Banerjee, Prof. Manjesh Hanawal* Spring 2023
Teaching Practicum for TAs | *Dept. of Computer Science and Engineering* Autumn 2022, Spring 2023
Software Systems Lab (Excellence in CSE TAship Award) | *Prof. Amitabha Sanyal* Autumn 2021
Computer Networks | *Prof. Vinay Ribeiro* Autumn 2022
Computer Programming and Utilization | *Prof. Kameswari Chebrolu* Spring 2021
Engineering Drawing | *Prof. Atul Sharma* Summer 2021